



Hosiery Knitting

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Background

1.1 Context

This document presents the modelling assumptions and data sources for **hosiery knitting**, representing the production of socks or similar small-diameter tubular knitted goods.

The model follows the same methodological principles applied in flat and circular knitting, ensuring coherence across all knitting process datasets.

1.2 Scope and functional unit

- **Functional unit:** 1 kg of hosiery fabric (average of mixed cotton and synthetic yarns).
- **System boundary:** Gate-to-gate hosiery knitting.
- **Included:**
 - Yarn losses, lubricants and related emissions
 - Electricity consumption (direct and indirect)
 - Machine depreciation (capital goods) and end-of-life (EoL)
 - Textile waste to treatment
- **Excluded:**
 - Fibre and yarn production (modelled upstream)
 - Sewing, dyeing, finishing and packaging
 - Factory building and auxiliary systems beyond indirect energy allocation

Technical Overview

2.1 Process characteristics

Hosiery knitting uses small-diameter single-cylinder circular knitting machines designed for automatic toe closure and optional linking.

Production rates, power consumption and material losses are highly dependent on the gauge, feed configuration and automation level.

Key benchmarks:

- **Output:** ~300–400 pairs/day per standard machine; premium multi-feed models can exceed 800 pairs/day (~33 pairs/h).
- **Machine weight:** typically **230 - 320 kg**.
- **Energy:** **0.5 - 1.2 kWh/kg** (machine electricity only).
- **Lubricant consumption:** **0.05–0.10 L/kg**, depending on lubrication system efficiency.
- **Yarn waste:** **2–5%**, depending on toe-closing system and yarn type.

Inventory Data

Category	Flow	Amount	Unit	Comment
Product	Hosiery fabric (average mix)	1	kg	Representative of mixed cotton/synthetic yarn composition.
Technosphere flows	Dtex Yarn, "e.g. Cotton" (yarn losses)	0.02	kg	Default 2% waste. Consistent with defect rates <2% reported for modern hosiery machines (Laursen 2007; van der Velden 2014; Sandin 2019; Accio, 2023).
	Lubricant, average	0.008	kg	Lubricant: 8 g kg ⁻¹ fabric "Sandin et al.2019 pp45 - The lubricant was assumed to correspond to 8% of the weight of the knit."
	Air emissions from lubricant	0.008	kg	Equal to lubricant input for simplification.
Electricity / heat	Electricity mix	1.2	kWh	Average SEC derived from literature and manufacturer data: circular knitting ≈ 0.8–1.2 kWh/kg (Maity et al., 2021; Lonati GK series ≈ 0.5 kW for 0.9 kg/h output).
Indirect energy	Compressors, HVAC, lighting, robots	4.46	kWh	Indirect uses (air compressors ~6 bar, cooling, illumination, auxiliaries). Harmonised with other knitting processes for methodological consistency.
Capital goods	Building Machine (market for building machine)	0.0004	kg/kg fabric	Based on 300 kg machine amortised over 720,000 kg lifetime output (see below).
Waste to treatment	EoL Machine (Steel)	0.00035	kg	85% of total mass (steel) distributed per kg of fabric.
	Recycling (Other materials)	0.00006	kg	15% of total mass (motors, electronics, plastics).
	Textile waste incineration (no credits)	0.02	kg	Seamless knitting eliminates cutting losses; total waste typically <5% (K&G Garment, 2025; AATCC in GIZ 2023).

4. Capital Goods (Machine)

4.1 Equipment data

Machine	Weight (kg)	Source
Lonati GE616DF / GE516DF (family)	300	Lonati / Santoni technical brochures (2023)

4.2 Allocation parameters

Scenario	Years	Hours/day	Days/year	Productivity (kg/h)	Lifetime output (kg)	Allocation (kg/kg fabric)
Average	15	16	300	10	720,000	0.0004

4.3 End-of-Life (EoL) allocation

Material	Share (%)	Weight (kg)	Allocation (kg/kg fabric)
Steel	85	255	0.00035
Other materials	15	45	0.00006

These allocations show the portion of the machine infrastructure "consumed" per kilogram of hosiery fabric.

4.4 Representation in ecoinvent

For the dataset "market for building machine":

- **Amount:** 1.4E-07 unit/kg
(0.0004 kg/kg ÷ 300 kg/machine)
- **Unit:** unit
- **Product:** building machine
- **Geography:** GLO

This approach mirrors that used for flat and circular knitting, aligning with EF and ecoinvent rules for infrastructure amortisation.

4.5 Building Infrastructure

The hosiery knitting process model includes an optional building infrastructure (pavilion) share as a modular and auditable input. The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m² of industrial hall.

4.5.1 Allocation principle (area amortisation over lifetime output).

Building infrastructure is attributed to 1 kg of knitted hosiery by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity (m}^2\text{/kg)} = \frac{A_{\text{alloc}} \text{ (m}^2\text{)}}{Q_{\text{lifetime}} \text{ (kg)}}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + yarn supply + maintenance space + internal logistics), and Q_{lifetime}

is the total knitted output produced during the reference service life under the chosen operating regime.

4.5.2 Technology rationale.

Hosiery machines are physically small relative to other knitting technologies (individual machine envelope below 1.5 m²), but are typically deployed in multi-machine cells with shared yarn feed, collective creel systems, and common aisles. The resulting cell area, normalised by throughput, is comparable to seamless knitting, placing hosiery in the same intensity band. Additionally, the low mass per article and relatively slow cycle time per gram of output sustain a moderate building intensity per kg.

4.5.3 Practical implementation.

In the hosiery knitting foreground unit process, we add an input: *building, hall* (unit = m²) with amount = 1.61E-05 m²/kg.

4.5.4 End-of-life of the building.

The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m²/kg input is provided.

4.5.5 Default value (this study). In the absence of site-specific A_{alloc} and Q_{lifetime} , the following reference intensity is applied, shared with the seamless technology group given comparable normalised throughput per allocated space:

■ Hosiery knitting: **1.61E-05 m²/kg**

Physical plausibility was cross-checked using machine envelope data from Lonati GE616DF3 (width 127.3 cm × depth 114.8 cm, envelope ≈ 1.46 m²) with a higher cell factor to account for multi-machine layout, shared yarn supply infrastructure, and aisles, yielding an allocated area consistent with the adopted default.

5. Indirect Energy Allocation

Indirect energy includes electricity consumed by **compressors, HVAC systems, lighting, and auxiliary robots** for automatic toe linking, packaging or maintenance.

As sub-metering is rarely available, the **total indirect energy of 4.46 kWh/kg** is used — the same reference adopted across all knitting processes for consistency.

This total can later be divided into subsystems using the default mid-range distribution:

Subsystem	Share (%)	Equivalent (kWh/kg)
Knitting machines	35	1.56
Compressors	25	1.12
HVAC	30	1.34
Lighting	6.5	0.29
Cleaning / auxiliaries	3.5	0.16

6. Waste Generation

Seamless hosiery knitting produces **minimal cutting waste**, in contrast to conventional cut-and-sew garments that lose 10–25% of material during trimming.

Industrial data (K&G Garment, 2025; GIZ 2023) show material utilisation above 95%, corresponding to **<5% total losses**.

The LCI therefore assumes **2–5% total yarn waste**, and **0.02 kg/kg textile waste** is modelled for treatment.

7. Key Assumptions Summary

Parameter	Average value	Comment
Electricity (machine)	1.2 kWh/kg	Lonati / literature average
Indirect energy (HVAC, compressors, etc.)	4.46 kWh/kg	Harmonised default for knitting
Lubricant	0.008 kg/kg	8 g/kg fabric typical maximum
Yarn loss	0.02 kg/kg	<2% defect rate typical
Machine mass allocation	0.0004 kg/kg	300 kg machine amortised
EoL (steel)	0.00035 kg/kg	85% share
Textile waste	0.02 kg/kg	<5% utilisation loss

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